

Solution (Queuing model)

1.

a) $L = 0\left(\frac{1}{16}\right) + 1\left(\frac{4}{16}\right) + 2\left(\frac{6}{16}\right) + 3\left(\frac{4}{16}\right) + 4\left(\frac{1}{16}\right) = 2$ which represents the average number of customers in the shop, including those getting their hair cut.

b)

n	# in queue	probability	product
0	0		
1	0		
2	0		
3	1	0.25	0.25
4	2	0.0625	0.125

$L_q = 0.375$ which represents the average number of customers in the shop waiting to get a haircut.

c) $E(\# \text{ customers being served}) = 1 \cdot P_1 + 2(P_2 + P_3 + P_4) =$
 $= \frac{4}{16} + 2\left(\frac{6}{16} + \frac{4}{16} + \frac{1}{16}\right) = \frac{13}{8}$

d) _____

$$W = \frac{L}{\lambda} = \frac{2}{4} = 0.5 = 30 \text{ minutes}$$

$$W_q = \frac{L_q}{\lambda} = \frac{0.375}{4} = 0.094 = 5.625 \text{ minutes}$$

These quantities mean that customers will be in the shop an average of half an hour, including the time to get a haircut, and will have to wait an average of 5.625 minutes before their haircut will begin.

e) $W - W_q = 0.5 - 0.094 = 0.406 \text{ hours} = 24.36 \text{ minutes}$

2.

$$\lambda_n = \frac{1}{2} \text{ for } n > 0 \text{ and } \mu_n = \begin{cases} \frac{1}{2} & \text{for } n=1 \\ 1 & \text{for } n \geq 2 \end{cases}$$

$$\begin{aligned} \text{(a) } P\{\text{next arrival before 1:00}\} &= 1 - e^{-1/2} = .393 \\ P\{\text{next arrival between 1:00 and 2:00}\} &= (1 - e^{-1/2}) - (1 - e^{-1/2}) = .239 \\ P\{\text{next arrival after 2:00}\} &= e^{-1/2} = .368 \end{aligned}$$

$$\text{(b) } P\{\text{next arrival between 1:00 and 2:00} \mid \text{no arrivals between 12:00 and 1:00}\} = 1 - e^{-1/2} = .393$$

$$\begin{aligned} \text{(c) } P\{\text{no arrivals between 1:00 and 2:00}\} &= \frac{(\lambda t)^0 e^{-\lambda t}}{0!} = e^{-1/2} = .607 \\ P\{\text{one arrival between 1:00 and 2:00}\} &= \frac{(\lambda t)^1 e^{-\lambda t}}{1!} = \frac{1}{2} \cdot e^{-1/2} = .303 \\ P\{\text{two or more arrivals between 1:00 and 2:00}\} &= 1 - e^{-1/2} - \frac{1}{2} e^{-1/2} \end{aligned}$$

$$\begin{aligned} \text{(d) } P\{\text{none served by 2:00}\} &= e^{-1} = .368 \\ P\{\text{none served by 1:10}\} &= e^{-1(1/6)} = .846 \\ P\{\text{none served by 1:01}\} &= e^{-1(1/60)} = .983 \end{aligned}$$